

H4: The rotation groups SU(2) and SO(3)	
4.1 warming up to SO(2) and U(1)	
4.1.1 rotations in the plane and group theory	
rotation matrix	consider a point in 2D space: $\begin{cases} x = r \cos \alpha, \\ y = r \sin \alpha. \end{cases}$ to move them counterclockwise over an angle θ, we have to add: $\begin{cases} x' = r \cos(\alpha + \theta) = r \cos \alpha \cos \theta - r \sin \alpha \sin \theta, \\ y' = r \sin(\alpha + \theta) = r \sin \alpha \cos \theta + r \cos \alpha \sin \theta. \end{cases}$ or in terms of old coords: $\begin{cases} x' = \cos \theta x - \sin \theta y, \\ y' = \sin \theta x + \cos \theta y. \end{cases}$ and in matrix: $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$ define: $R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$
active vs passive pov	<u>active</u> you rotate the vector in a fixed coordinate frame > action is implemented by acting $R(\theta)$ on the vector <u>passive</u> you change the basis according to: $\{e_x, e_y\} \rightarrow \{R(\theta)^{-1}e_x, R(\theta)^{-1}e_y\}$ ie: you change the frame instead of the vector
group of $R(\theta)$	notice that the collection of matrices $\{R(\theta) \theta \in [0, 2\pi]\}$ forms a group with mult the matrix mult > this is because it has the properties: - The identity obviously corresponds to $R(\theta = 0) = \mathbb{1}_2$. Geometrically this means that a 'rotation' with an angle of 0 is equivalent to not rotating at all. Makes sense. - For every rotation $R(\theta)$ there is an inverse rotation $R(2\pi - \theta) = R(-\theta)$: $R(\theta)R(2\pi - \theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \quad (4.11)$ $= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = R(2\pi - \theta)R(\theta) = \mathbb{1}_2. \quad (4.12)$ Notice that this is equivalent to saying that the rotation matrix $R(\theta)$ is orthogonal: $R(\theta)^{-1} = R(\theta)^T = R(2\pi - \theta)$. - The matrix multiplication is by construction associative.
def: special orthogonal group SO(2)	This is the Lie group: $SO(2) := \left\{ \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \mid \theta \in [0, 2\pi) \right\}$
defining representation	Many Lie groups are defined through a representation > call this representation the defining representation
linear Lie group	= Lie groups that can be defined via a matrix representation
def: special orthogonal group SO(2)	We can also define the 2D rotation group as: $SO(2) := \{O \in \mathbb{R}^{2 \times 2} \mid OO^T = O^T O = \mathbb{1}, \det O = 1\}$ where orthogonal refers to the fact that $O^T = O^{-1}$ and special since $\det O = 1$
def: orthogonal group O(2)	this is the matrix Lie group: $O(2) := \{O \in \mathbb{R}^{2 \times 2} \mid OO^T = O^T O = \mathbb{1}\}$

meaning of O(2)	because of: $\det(AB) = \det(A)\det(B)$ and $\det(A^T) = \det(A)$ then the determinant of all matrices in O(2) are +-1 > if they are +1 they are in SO(2) > for det=-1 we find that they represent reflection around the y-axis
SO(2) an abelian group	rotating over an angle θ_1 followed by rotation over θ_2 is exactly the same as rotating over θ_2 first and then θ_1 > thus SO(2) is an abelian group: $\begin{pmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{pmatrix} \begin{pmatrix} \cos \theta_2 & -\sin \theta_2 \\ \sin \theta_2 & \cos \theta_2 \end{pmatrix} = \begin{pmatrix} \cos \theta_2 & -\sin \theta_2 \\ \sin \theta_2 & \cos \theta_2 \end{pmatrix} \begin{pmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{pmatrix} \\ = \begin{pmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) \end{pmatrix}.$
th: $SO(2) \simeq U(1)$	The rotation group SO(2) is isomorphic to U(1): $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \in SO(2) \rightarrow e^{i\theta} \in U(1), \quad \forall \theta \in [0, 2\pi).$
4.1.2 irreducible representations U(1) and SO(2)	
irreps of $U(1) \simeq SO(2)$	The irreps of an abelian group are 1D and unitary ie: they are homomorphisms $\varphi: U(1) \rightarrow U(1)$ and since $\varphi(1)=1$ we have irreps of the form: $\varphi_n(e^{i\theta}) = e^{in\theta}, \quad n \in \mathbb{Z}.$ thus for every integer n we have an irrep
4.2 definitions and parametrizations of SO(3) and SU(2)	
4.2.1 definitions	
def: orthogonal group O(3)	= group of real 3x3 orthogonal matrices: $O(3) := \{M \in \mathbb{R}^{3 \times 3} MM^T = M^T M = \mathbb{1}\}$ From the fact that $\det(AB) = \det(A)\det(B)$ and $\det(A^T) = \det(A)$, we know $\det = \pm 1$
def: special orthogonal group SO(3)	= group of real 3x3 orthogonal matrices with unit determinant: $SO(3) := \{M \in \mathbb{R}^{3 \times 3} MM^T = M^T M = \mathbb{1}, \det M = 1\} < O(3).$ > 3D rotations
def: unitary group U(2)	= group of complex unitary 2x2 matrices: $U(2) := \{U \in \mathbb{C}^{2 \times 2} UU^\dagger = U^\dagger U = \mathbb{1}\}$ from $\det(AB) = \det(A)\det(B)$ and $\det(A^\dagger) = \overline{\det(A)}$ we know that the det is a phase $e^{i\phi}$
def: special unitary group SU(2)	= group of complex unitary 2x2 matrices with unit determinant: $SU(2) := \{U \in \mathbb{C}^{2 \times 2} UU^\dagger = U^\dagger U = \mathbb{1}, \det U = 1\} < U(2).$
4.2.2 parametrizations	
<u>parametrization of SO(3)</u>	
1: rotation in 3D space	The most obvious way is just rotating over the x, y or z axis: $R(\theta_x, \theta_y, \theta_z) = R_z(\theta_z)R_y(\theta_y)R_x(\theta_x),$ with: $R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix}$ $R_y(\theta) = \begin{pmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{pmatrix}$ $R_z(\theta) = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$ >> these do NOT commute ie: the order of xyz matters > furthermore, these rotations are passive

2: axis of rotation	rotations always leave one axis fixed, the <i>axis of rotation</i> > thus: choose an axis defined by unit vector $\mathbf{n}$ and an angle $\theta \in [0, \pi]$ > denote the rotation as $R_{\mathbf{n}}(\theta)$
3: Euler's angles	These rotations are done in three steps: 1: rotate over the z-axis 2: rotate over the new x-axis 3: rotate over the new z-axis: $R(\psi, \theta, \varphi) = \begin{pmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$ >> these are passive
<u>parametrization of SU(2)</u>	
1: Pauli matrices	$U = \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix}, \quad \alpha, \beta \in \mathbb{C}, \quad \alpha ^2 + \beta ^2 = 1. \quad (4.45)$ Writing $\alpha = \alpha_1 + i\alpha_2$, $\beta = \beta_1 + i\beta_2$ with α_i and β_i real allows us to also write every matrix in $SU(2)$ in terms of the by now familiar Pauli matrices: $\begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} = \alpha_1 \mathbb{1}_2 + i\beta_2 \sigma_X + i\beta_1 \sigma_Y + i\alpha_2 \sigma_Z, \quad \alpha_1, \alpha_2, \beta_1, \beta_2 \in \mathbb{R}, \quad (4.46)$ $\sigma_X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (4.47)$
2: matrix exponentials	write $t_i := \sigma_i/2$ and $\sigma_{\mathbf{n}} = n_x t_x + n_y t_y + n_z t_z$ with $\mathbf{n}$ a real unit vector > every matrix in $SU(2)$ can be written as: $R_{\vec{n}}(\theta) = e^{i\theta \sigma_{\vec{n}}} = \cos\left(\frac{\theta}{2}\right) \mathbb{1}_2 + 2i \sin\left(\frac{\theta}{2}\right) \sigma_{\vec{n}}, \quad \theta \in [0, 2\pi]$
4.3 the Lie algebras $\mathfrak{so}(3)$ and $\mathfrak{su}(2)$	
infinitesimal rotations	Let us now rotate over a small angle $\delta\theta$, then due to Taylor: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \delta\theta & \sin \delta\theta \\ 0 & -\sin \delta\theta & \cos \delta\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \delta\theta \\ 0 & -\delta\theta & 1 \end{pmatrix} + \mathcal{O}(\delta\theta^2) = \mathbb{1}_3 + i\delta\theta \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}}_{L_x} + \mathcal{O}(\delta\theta^2).$ similarly we find in y and z: $L_y = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \quad L_z = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ thus, any small rotation in 3D can be written as: $\mathbb{1}_3 + i\delta\theta (n_x L_x + n_y L_y + n_z L_z)$ now L_x, L_y, L_z do not commute > it matters in which direction you rotate first > we have the relation: $[L_i, L_j] = i \sum_k \varepsilon_{ijk} L_k,$ with: $\varepsilon_{ijk} = \begin{cases} +1 & \text{if } (i, j, k) \text{ is } (1, 2, 3), (2, 3, 1), \text{ or } (3, 1, 2), \\ -1 & \text{if } (i, j, k) \text{ is } (3, 2, 1), (1, 3, 2), \text{ or } (2, 1, 3), \\ 0 & \text{if } i = j, \text{ or } j = k, \text{ or } k = i \end{cases}$

Lie bracket	= a function $[-, -] : \mathfrak{g} \times \mathfrak{g} \mapsto \mathfrak{g}$⁵ that is: 1. bilinear $[\lambda a + b, c + \lambda' d] = \lambda[a, c] + \lambda\lambda'[a, d] + [b, c] + \lambda'[b, d]$, 2. antisymmetric $[a, b] = -[b, a]$, 3. and satisfies the Jacobi identity $[a, [b, c]] + [b, [c, a]] + [c, [a, b]] = 0,$ 0 being the zero vector.
Lie algebra	vector space that is endowed with a Lie Bracket
generators of $\mathfrak{so}(3)$	The matrices L_1, L_2, L_3 form a 3D real vector space > together with the commutator $[.,.]$ as Lie bracket, they form a Lie algebra > these matrices are the basis vectors of the Lie algebra called $\mathfrak{so}(3)$ Now: $\mathfrak{so}(3)$ is a real vector space > however, L_i are complex matrices > this is still valid, since we only consider real combinations of these generators
Lie algebra $\mathfrak{su}(2)$	We can do the same exercise for $SU(2)$: $R_{\vec{n}}(\theta) = e^{i\theta\sigma_{\vec{n}}} = \cos\left(\frac{\theta}{2}\right)\mathbb{1}_2 + 2i\sin\left(\frac{\theta}{2}\right)\sigma_{\vec{n}}, \quad \theta \in [0, 2\pi[.$ using Taylor, we can write for small angles: $R_{\vec{n}}(\delta\theta) \approx \mathbb{1}_2 + i\delta\theta \left(n_x \frac{\sigma_x}{2} + n_y \frac{\sigma_y}{2} + n_z \frac{\sigma_z}{2} \right)$ thus, infinitesimal rotations of $SU(2)$ are generated by: $L_x = \frac{\sigma_x}{2} = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$ $L_y = \frac{\sigma_y}{2} = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ $L_z = \frac{\sigma_z}{2} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ For which the commutation relations are exactly the same as for $\mathfrak{so}(3)$: $\left[\frac{\sigma_i}{2}, \frac{\sigma_j}{2} \right] = i \sum_k \varepsilon_{ijk} \frac{\sigma_k}{2}$ This means $\mathfrak{su}(2)$ is isomorphic to $\mathfrak{so}(3)$ > these are two distinct irreducible representations of the same Lie algebra $\mathfrak{su}(2) \simeq \mathfrak{so}(3)$
4.4 topological aspects $SU(2)$ and $SO(3)$	
Topological $SU(2)$	we know $SU(2)$ is parametrized by: $U = \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix}, \quad \alpha, \beta \in \mathbb{C}, \quad \alpha ^2 + \beta ^2 = 1. \quad \text{with } \alpha = \alpha_1 + i\alpha_2, \beta = \beta_1 + i\beta_2 \text{ with } \alpha_i \text{ and } \beta_i \text{ real}$ > defines a 3D sphere S^3 > topologically speaking, S^3 is compact
Topological $SO(3)$	$SO(3)$ can be parametrized by $R_{\vec{n}}(\theta)$ with: $\vec{n}$ a normalized vector defining the axis of rotation θ an angle between 0 and π > $SO(3)$ is a ball with radius π
> difference between $SU(2)$ and $SO(3)$	We have seen that the Lie Algebras of $SO(3)$ and $SU(2)$ are isomorphic > locally speaking, $SO(3)$ and $SU(2)$ are identical > however looking globally, aka topologically, we see that $SU(2)$ and $SO(3)$ define different objects

4.5 irreducible representations SU(2) and SO(3)

4.5.1 irreducible representations $\mathfrak{sl}(2)$

ladder operators

For $\mathfrak{su}(2)$ we can construct the ladder operators:

$$L_{\pm} := L_1 \pm iL_2, \quad L_0 := 2L_3.$$

which have the property:

$$[L_0, L_{\pm}] = \pm 2L_{\pm},$$

$$[L_+, L_-] = L_0.$$

$L_{\pm}$ are not Hermitian, they rather satisfy $L_{\pm}^{\dagger} = L_{\mp}$

Define the quadratic Casimir operator:

$$\vec{L}^2 := L_x L_x + L_y L_y + L_z L_z = \frac{1}{2}(L_+ L_- + L_- L_+) + \frac{L_0^2}{4}$$

this has the property, that for all generators L_i :

$$[\vec{L}^2, L_i] = 0,$$

furthermore:

$$[AB, C] = A[B, C] + [A, C]B \text{ and } [A, BC] = B[A, C] + [A, B]C$$

We also have the relation:

$$L_{\pm} L_{\mp} = \vec{L}^2 - L_3^2 \pm L_3.$$

complexification of $\mathfrak{su}(2)$: $\mathfrak{sl}(2)$

$L_{\pm}$ are complex, thus they define a complex Lie algebra

> call this the complexification of $\mathfrak{su}(2)$ denoted by $\mathfrak{sl}(2)$

Highest weight representations

We will calculate the irreps of $\mathfrak{sl}(2)$ rather than $\mathfrak{su}(2)$

> however: these can be proven to be the same

Both L_3 and L^2 are Hermitian operators

> they have real eigenvalues, denoted by resp m and l

> for L^2 the eigenvalues are non-negative, thus we can write them as $l(l+1)$

> there exists a simultaneous eigenbasis for L_3 and L^2 , denoted by $\{|l, m\rangle\}$, where:

$$\vec{L}^2 |l, m\rangle = l(l+1) |l, m\rangle$$

$$L_3 |l, m\rangle = m |l, m\rangle.$$

these eigenvectors are orthonormal:

$$\langle l, m | l', m' \rangle = \delta_{l,l'} \delta_{m,m'}$$

the eigenvalues l are called weights

> from the commutation relations we then have:

$$L_3 L_{\pm} |l, m\rangle = (L_{\pm} L_3 \pm L_{\pm}) |l, m\rangle = (m \pm 1) L_{\pm} |l, m\rangle$$

this is why they are called ladder operations

nl: given an eigenvector $|l, m\rangle$ acting $L_{\pm}$ will get you an eigenvector with eigenvalue $m \pm 1$

The highest weight has the functionality that:

1: start from an eigenvector with maximal eigenvalue M , the highest weight

> this eigenvalue is annihilated by L_+ :

$$L_+ |l, M\rangle = 0$$

2: sequentially act L_- on this highest weight factor

> we get a closed string of vectors under action of $\mathfrak{sl}(2)$:

$$\{|l, M\rangle, |l, M-1\rangle, |l, M-2\rangle, \dots\}$$

> forms a representation under $\mathfrak{sl}(2)$

3: l is a constant in a given representation

nl: l is the eigenvalue of L^2

> L^2 commutes with the ladder operator, thus they don't change this eigenvalue

4: to be finite-dimensional, there exists some M' corresponding to eigenvector $|l, M'\rangle$ which is annihilated by L_- :

$$L_- |l, M'\rangle = 0$$

> the string S_l only has a finite number of vectors:

$$S_l := \{|l, M\rangle, |l, M-1\rangle, |l, M-2\rangle, \dots, |l, M'\rangle\}$$

> constitutes a finite-dimensional representation

> the dimension of the irrep $V_l := \text{span}_{\mathbb{C}}(S_l)$ is then of course $\dim_{\mathbb{C}} V_l = M - M' + 1$

relation l and M	We can now relate M to l as follows: $\begin{aligned} L_- L_+ l, M\rangle &= \left(\tilde{L}^2 - L_3^2 - L_3\right) l, M\rangle \\ &= (l(l+1) - M^2 - M) l, M\rangle \\ &= 0. \end{aligned}$ Here we used (4.71). Hence, $l(l+1) = M(M+1).$																									
relation l and M'	From (4.84) on the other hand it follows that $\begin{aligned} L_+ L_- l, M'\rangle &= \left(\tilde{L}^2 - L_3^2 + L_3\right) l, M'\rangle \\ &= (l(l+1) - M'^2 + M') l, M'\rangle \\ &= 0. \end{aligned}$ We find: $l(l+1) = M'(M'-1).$																									
relation M and M'	$M(M+1) = M'(M'-1) \iff M' = -M$																									
> conclusion irrep of $\mathfrak{sl}(2)$	All finite dimensional irreps $\mathfrak{sl}(2)$ are labelled by one integer l > dimension of that irrep is: $\dim_{\mathbb{C}} V_l = M - M' + 1 = 2l + 1$ call l = spin quantum number m = magnetic quantum number <table><tr><th>Dimension</th><th>l</th><th>l(l+1)</th><th>d</th><th>Common names</th></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>Trivial, singlet, scalar</td></tr><tr><td>2</td><td>$\frac{1}{2}$</td><td>$\frac{3}{4}$</td><td>2</td><td>Fundamental, defining, doublet, spinor</td></tr><tr><td>3</td><td>1</td><td>2</td><td>3</td><td>Adjoint, triplet, vector</td></tr><tr><td>$\vdots$</td><td>$\vdots$</td><td>$\vdots$</td><td>$\vdots$</td><td>$\vdots$</td></tr></table>	Dimension	l	l(l+1)	d	Common names	1	0	0	1	Trivial, singlet, scalar	2	$\frac{1}{2}$	$\frac{3}{4}$	2	Fundamental, defining, doublet, spinor	3	1	2	3	Adjoint, triplet, vector	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
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$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$																						
4.5.2 computing matrix representations of $\mathfrak{sl}(2)$																										
matrix representation of L^2 and L_3	the eigenvectors are orthonormal: $\langle l, m l', m' \rangle = \delta_{l,l'} \delta_{m,m'}$ thus we have for L^2: $\langle l', m' \tilde{L}^2 l, m \rangle = l(l+1) \delta_{l,l'} \delta_{m,m'}.$ and for L_3 $\langle l', m' L_3 l, m \rangle = m \delta_{l,l'} \delta_{m,m'}$																									
matrix representation of $L_{\pm}$	We know that $L_{\pm} l, m\rangle$ are proportional to $l, m \pm 1\rangle$ > we can thus write: $L_{\pm} l, m\rangle = N_{\pm} l, m \pm 1\rangle$ we can find: $\begin{aligned} N_{\mp} ^2 &= \langle l, m L_{\mp}^{\dagger} L_{\mp} l, m \rangle \\ &= \langle l, m L_{\pm} L_{\mp} l, m \rangle \\ &= (l(l+1) - m(m \mp 1)) \end{aligned}$ thus: $N_{\pm} = \sqrt{l(l+1) - m(m \pm 1)}.$ we then have for L_+: $\begin{aligned} \langle l', m' L_+ l, m \rangle &= N_+ \delta_{l,l'} \delta_{m+1,m'} \\ &= \sqrt{l(l+1) - m(m+1)} \delta_{l,l'} \delta_{m+1,m'} \end{aligned}$ and L_-: $\begin{aligned} \langle l', m' L_- l, m \rangle &= N_- \delta_{l,l'} \delta_{m-1,m'} \\ &= \sqrt{l(l+1) - m(m-1)} \delta_{l,l'} \delta_{m-1,m'} \end{aligned}$																									

4.5.4 irreducible representations of SU(2), the exponential map and the BCH formula

irreps of SU(2)	The irreps of SU(2) is the exponential map of $\mathfrak{su}(2)$
def: matrix exponential	Via Taylor: $\exp(M) := \sum_{k=0}^{\infty} \frac{M^k}{k!}$
def: BCH formula	given two matrices A and B: $\exp(A) \exp(B) = \exp(C),$ $C = A + B + \frac{1}{2}[A, B] + \frac{1}{12}[A, [A, B]] - \frac{1}{12}[B, [A, B]] + \dots$ or simply in second order: $C = A + B + \frac{1}{2}[A, B]$ If A and B commute we simply have: $\exp(A) \exp(B) = \exp(A + B)$

4.6 tensor product representations $\mathfrak{sl}(2)$ and Clebsch-Gordan coefficients

uncoupled basis for $\mathfrak{sl}(2)$	suppose we have two sets of $\mathfrak{su}(2)$ generators and their quadratic Casimirs > denote this using superscript (1) or (2) > the commutators are: $[L_i^{(1)}, L_j^{(1)}] = i \sum_k \epsilon_{ijk} L_k^{(1)} \quad [L_i^{(2)}, L_j^{(2)}] = i \sum_k \epsilon_{ijk} L_k^{(2)}$ $[L_i^{(1)}, \overrightarrow{L^{(1)}}^2] = 0, \quad [L_i^{(2)}, \overrightarrow{L^{(2)}}^2] = 0.$ we know that a $2l_1+1$ dimensional irrep of the 1st set of generators is given by basis states: $V_{l_1} := \{ l_1, m_1\rangle m_1 = l_1, l_1 - 1, \dots, -l_1\}$ for which: $L_3^{(1)} l_1, m_1\rangle = m_1 l_1, m_1\rangle,$ $\overrightarrow{L^{(1)}}^2 l_1, m_1\rangle = l_1(l_1 + 1) l_1, m_1\rangle$ similar for $2l_2+1$ dimensional irrep of the 2nd set of generators that's given by basis states: $V_{l_2} := \{ l_2, m_2\rangle m_2 = l_2, l_2 - 1, \dots, -l_2\}$ for which: $L_3^{(2)} l_2, m_2\rangle = m_2 l_2, m_2\rangle,$ $\overrightarrow{L^{(2)}}^2 l_2, m_2\rangle = l_2(l_2 + 1) l_2, m_2\rangle$ Now we want to form a tensor product representation of $\mathfrak{sl}(2)$ > first define the operators: $L_{1,i} := L_i^{(1)} \otimes \mathbb{1} \quad \tilde{L}_1^2 := \overrightarrow{L^{(1)}}^2 \otimes \mathbb{1}$ $L_{2,i} := \mathbb{1} \otimes L_i^{(2)} \quad \tilde{L}_2^2 := \mathbb{1} \otimes \overrightarrow{L^{(2)}}^2$ these commute: $[L_{1,i}, L_{2,i}] = 0 \quad [\tilde{L}_1^2, L_{2,i}] = 0$ $[L_{1,i}, \tilde{L}_2^2] = 0 \quad [\tilde{L}_1^2, \tilde{L}_2^2] = 0$ thus there is a basis in which the operators $\{L_{1,3}, L_{2,3}, \tilde{L}_1^2, \tilde{L}_2^2\}$ can be simultaneously diagonalised > the simultaneous eigenbasis is the tensor product basis: $\{ l_1, m_1\rangle l_2, m_2\rangle m_1 = l_1, l_1 - 1, \dots, -l_1, m_2 = l_2, l_2 - 1, \dots, -l_2\}$ this basis spans a Hilbert space of dimension $(2l_1+1)(2l_2+1)$ > on the basis states the operators act according to: $L_{1,3} l_1, m_1\rangle l_2, m_2\rangle = m_1 l_1, m_1\rangle l_2, m_2\rangle,$ $L_{2,3} l_1, m_1\rangle l_2, m_2\rangle = m_2 l_1, m_1\rangle l_2, m_2\rangle,$ $\tilde{L}_1^2 l_1, m_1\rangle l_2, m_2\rangle = l_1(l_1 + 1) l_1, m_1\rangle l_2, m_2\rangle$ $\tilde{L}_2^2 l_1, m_1\rangle l_2, m_2\rangle = l_2(l_2 + 1) l_1, m_1\rangle l_2, m_2\rangle$ call this basis the <i>uncoupled basis</i> > forms the basis of the tensor product representation denoted as: $l_1 \otimes l_2$
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coupled basis for $\mathfrak{sl}(2)$	define the operators: $L_i := L_{1,i} + L_{2,i}$, $\vec{L}^2 := \vec{L}_1^2 + \vec{L}_2^2$. these satisfy again: $[L_i, L_j] = i \sum_k \varepsilon_{ijk} L_k,$ $[L_i, \vec{L}^2] = 0.$ Apply the results from the previous section: irreps of these operations have a basis: $\{ l, m\rangle m = l, l-1, \dots, -l\}$ the action of the operators is given by: $L_3 l, m\rangle = m l, m\rangle$ $\vec{L}^2 l, m\rangle = l(l+1) l, m\rangle$ this basis is called the <i>coupled basis</i>
th: tensor product of l_1 and l_2	Given spin l_1 and l_2 representations of $\mathfrak{sl}(2)$, their tensor product decomp as a direct sum as: $l_1 \otimes l_2 \simeq \bigoplus_{l= l_1-l_2 }^{l_1+l_2} l.$ $\simeq$ means 'up to a basis transformation'.
4.6.2 computing Clebsch-Gordan coefficients	
Clebsch-Gordan coefficients	= coefficients C in: $l, m\rangle = \sum_{m_1+m_2=m} C_{m_1 m_2 m}^{l_1 l_2 l} l_1, m_1\rangle l_2, m_2\rangle$ where we only sum over the allowed values of m: $m \stackrel{!}{=} m_1 + m_2$ using the notation: $l_1, m_1\rangle \otimes l_2, m_2\rangle \equiv l_1, m_1\rangle l_2, m_2\rangle \equiv l_1, m_1; l_2, m_2\rangle$ the CG's are found via: $C_{m_1 m_2 m}^{l_1 l_2 l} = \langle l_1, m_1; l_2, m_2 l, m \rangle$
CG's form unitary basis transformation	The CG coefficients form a unitary basis transformation > the orthonormality and completeness read: $\sum_{m_1, m_2} \langle l_1, m_1; l_2, m_2 l, m \rangle \langle l_1, m_1; l_2, m_2 l', m' \rangle = \delta_{l, l'} \delta_{m, m'},$ $\sum_{l, m} \langle l_1, m_1; l_2, m_2 l, m \rangle \langle l_1, m'_1; l_2, m'_2 l, m \rangle = \delta_{m_1, m'_1} \delta_{m_2, m'_2}.$
4.6.3 coupling of two spinors	
example: coupling of 2 spinors	Consider the coupling of two spin 1/2 particles > we are looking for a basis transformation between the uncoupled direct product basis: $\left \frac{1}{2}, \frac{1}{2} \right\rangle \left \frac{1}{2}, \frac{1}{2} \right\rangle, \left \frac{1}{2}, \frac{1}{2} \right\rangle \left \frac{1}{2}, -\frac{1}{2} \right\rangle, \left \frac{1}{2}, -\frac{1}{2} \right\rangle \left \frac{1}{2}, \frac{1}{2} \right\rangle, \left \frac{1}{2}, -\frac{1}{2} \right\rangle \left \frac{1}{2}, -\frac{1}{2} \right\rangle$ denoted by: $ \uparrow\uparrow\rangle, \uparrow\downarrow\rangle, \downarrow\uparrow\rangle, \downarrow\downarrow\rangle$ and the coupled basis given by the basis states: $ 1, 1\rangle, 1, 0\rangle, 1, -1\rangle, 0, 0\rangle$ Now: the constraint $m \stackrel{!}{=} m_1 + m_2$ results in: $ 1, 1\rangle = \uparrow\uparrow\rangle$ $ 1, -1\rangle = \downarrow\downarrow\rangle$ > CG coefficient is +1 Now to find $1, 0\rangle$ we can use $\langle l', m' L_- l, m \rangle = N_- \delta_{l, l'} \delta_{m-1, m'}$ $= \sqrt{l(l+1) - m(m-1)} \delta_{l, l'} \delta_{m-1, m'}$ nl: $L_- 1, 1\rangle = \sqrt{1(1+1) - 1(1-1)} 1, 0\rangle$ $= \sqrt{2} 1, 0\rangle.$ but since $L_- = L_{1-} + L_{2-}$ we can write: $L_- 1, 1\rangle = (L_{1-} + L_{2-}) \uparrow\uparrow\rangle$ $= (L_{1-} \uparrow\rangle) \uparrow\rangle + \uparrow\rangle (L_{2-} \uparrow\rangle)$ $= \downarrow\uparrow\rangle + \uparrow\downarrow\rangle.$

example: coupling of 2 spinors	these two results give us: $ 1, 0\rangle = \frac{1}{\sqrt{2}} (\uparrow\downarrow\rangle + \downarrow\uparrow\rangle)$ and due to $m=m_1+m_2$, the state $0,0\rangle$ can only be of the form: $ 0, 0\rangle = \alpha \uparrow\downarrow\rangle + \beta \downarrow\uparrow\rangle$ due to orthogonality $\langle 1,0 0,0\rangle = 0$ we find: $\begin{aligned}\langle 1, 0 0, 0\rangle &= \frac{1}{\sqrt{2}} (\alpha \langle \uparrow\downarrow \uparrow\downarrow\rangle + \alpha \langle \uparrow\downarrow \downarrow\uparrow\rangle + \beta \langle \uparrow\downarrow \uparrow\downarrow\rangle + \beta \langle \uparrow\downarrow \downarrow\uparrow\rangle) \\ &= \frac{1}{\sqrt{2}} (\alpha \cdot 1 + \alpha \cdot 0 + \beta \cdot 0 + \beta \cdot 1) \\ &= \frac{1}{\sqrt{2}} (\alpha + \beta)\end{aligned}$ thus $\alpha=-\beta$, and due to normalization: $ \alpha ^2 + \beta ^2 \stackrel{!}{=} 1,$ we find: $ 0, 0\rangle = \frac{1}{\sqrt{2}} (\uparrow\downarrow\rangle - \downarrow\uparrow\rangle)$ thus the basis transformation is: $\begin{aligned} 1, 1\rangle &= \uparrow\uparrow\rangle, & \uparrow\uparrow\rangle &= 1, 1\rangle, \\ 1, 0\rangle &= \frac{1}{\sqrt{2}} (\uparrow\downarrow\rangle + \downarrow\uparrow\rangle), & \uparrow\downarrow\rangle &= \frac{1}{\sqrt{2}} (1, 0\rangle + 0, 0\rangle) \\ 1, -1\rangle &= \downarrow\downarrow\rangle, & \downarrow\uparrow\rangle &= \frac{1}{\sqrt{2}} (1, 0\rangle - 0, 0\rangle) \\ 0, 0\rangle &= \frac{1}{\sqrt{2}} (\uparrow\downarrow\rangle - \downarrow\uparrow\rangle), & \downarrow\downarrow\rangle &= 1, -1\rangle.\end{aligned}$
4.7 spherical harmonics	
4.7.1 motivation	
wave equation	the Helmholtz equation: $\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) f(\vec{r}, t) = 0.$ or after separation of variables: $f=T(t)\psi(r): \nabla^2\psi = -k^2\psi$ $\frac{d^2T}{dt^2} = -k^2 c^2 T.$
electrostatics	Maxwell's equations have $SO(3)$ symmetry nl: we have for a potential V: $\nabla^2 V = 0.$ > this is the Laplace equation > asking whether we can express solutions in a natural basis which respects $SO(3)$ symmetry leads to spherical harmonics
classical gravity	We know $\nabla g=0$, thus via a potential $g=-\nabla V$: $\nabla^2 V = 0$
particle in central potential	consider a central potential: $\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 + V(r)$ now: $\hat{H}$ commutes with L^2 and one component of L, conventionally L_y > we then check: $[L_i, L_j] = i\epsilon_{ijk} L_k$ $[L^2, L_i] = 0$ hence we can simultaneously diagonalise these three operators ex: Hydrogen atom > spectrum $n, l, m\rangle$ where $n=1,2,\dots$ $l=0,1,\dots,n-1$ and $m = -l, -l+1, \dots, l$ > then the energy: $E_n = -\frac{\mu e^4}{2\hbar^2 n^2}$ the states $\{ n, l, -l\rangle, n, l, -l+1\rangle, \dots, n, l, l\rangle\}$ form a $2l+1$-dimensional multiplet > or thus an irreducible representation of $SO(3)$

4.7.2 definitions and properties

spherical harmonics

The spherical harmonics form a complete set of basis states of the Hilbert space of square-integrable functions on the sphere $L^2(S^2)$:

$$f(\theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l f_l^m Y_l^m(\theta, \varphi) \quad (\text{Completeness spherical harmonics})$$

this basis is orthonormal:

$$\int_0^\pi \int_0^{2\pi} Y_l^m \bar{Y}_{l'}^{m'} d\Omega = \delta_{l,l'} \delta_{m,m'} \quad (\text{Orthonormality spherical harmonics})$$

Using these identities the Fourier coefficients can be expressed as

$$f_l^m = \int f(\theta, \varphi) \bar{Y}_l^m(\theta, \varphi) d\Omega \quad (4.244)$$

Under a rotation $R : (\theta, \varphi) \mapsto (\theta', \varphi')$ we have

$$Y_l^m(\theta', \varphi') = \sum_{m'=-l}^l D_{m'm}^l Y_l^{m'}(\theta, \varphi) \quad (4.245)$$

where $D_{m'm}^l \equiv D_{m'm}^l(R)$ depends on the rotation R and forms for a fixed value of $l = 0, 1, \dots$ a $2l + 1$ dimensional **irreducible** representation of the rotation group:

$$\sum_{m'=-l}^l D_{mm'}^l(R) D_{m'm''}^l(R') = D_{mm''}^l(R \cdot R') \quad (4.246)$$

$$D^l(R) D^{l'}(R') = D^l(R \cdot R') \quad (4.247)$$